

09_Design_System.md

Part 3 — Motion, Responsive System & Design Governance

Motion Design Philosophy

Animations should:

- Improve usability
- Provide visual feedback
- Guide user attention
- Communicate system state

Animations must never exist purely for decoration.

Motion Principles

- Fast
 - Smooth
 - Purposeful
 - Consistent
 - Interruptible
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Animation Categories

Micro Interactions

Examples

- Button Hover
 - Button Press
 - Toggle Switch
 - Checkbox
 - Input Focus
 - Card Hover
-

Page Transitions

Examples

- Fade
- Slide
- Scale
- Shared Element Transition

Rules

- Keep transitions consistent.
 - Avoid excessive animation duration.
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Loading Animations

Components

- Skeleton Loader
- Progress Indicator
- Spinner
- Progress Bar

Rules

- Always provide visual feedback.
 - Prevent layout shifts.
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Scroll Animations

Used On

- Landing Page
- Hero Section
- Feature Sections
- Statistics
- Testimonials

Rules

- Trigger once when appropriate.
 - Maintain smooth performance.
 - Never block user interaction.
-

Responsive Design System

Desktop

Minimum Width

1280px

Layout

- Sidebar Navigation
 - Multi-column Grid
 - Expanded Analytics
 - Full Charts
-

Tablet

Width

768px – 1279px

Layout

- Collapsible Sidebar
 - Adaptive Grid
 - Medium Cards
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Mobile

Maximum Width

767px

Layout

- Bottom Navigation
 - Single-column Layout
 - Floating Action Button
 - Full-width Cards
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Responsive Rules

- Mobile First Development
 - Flexible Grid Layout
 - Responsive Typography
 - Responsive Spacing
 - Responsive Images
-

Breakpoint System

Device	Width
Mobile	0-767px
Tablet	768-1279px
Desktop	1280px+

Empty States

Every empty state should contain:

- Illustration
- Title
- Description
- Primary CTA
- Optional Secondary CTA

Examples

- Empty Portfolio
 - Empty Watchlist
 - No Courses
 - No Competitions
 - No Notifications
-

Error States

Every error screen should include:

- Error Title
- Short Description
- Recovery Action
- Retry Button

- [Navigation Back](#)
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Loading States

Required For

- Dashboard
- Market
- Portfolio
- AI Coach
- Learning
- Leaderboards

Preferred Component

- Skeleton Loader
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Design Quality Standards

Every screen must satisfy:

- Visual consistency
 - Accessibility compliance
 - Responsive behaviour
 - Performance optimization
 - Token-based styling
 - Component reuse
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Design Review Checklist

Visual

- Typography follows hierarchy
 - Correct spacing
 - Correct colors
 - Consistent icons
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UX

- Navigation is intuitive
 - CTA placement is clear
 - Important information prioritized
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Accessibility

- Keyboard navigation
 - Contrast compliance
 - Focus indicators
 - Screen reader support
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Performance

- Optimized assets
 - Lazy-loaded images
 - Minimal layout shifts
 - Smooth animations
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Documentation Rules

Every component must include:

- Name
 - Purpose
 - Variants
 - States
 - Properties
 - Accessibility Notes
 - Usage Guidelines
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Versioning

Major Version

- Structural changes

Minor Version

- New components

Patch Version

- Bug fixes
 - Documentation updates
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Governance

The Design System is the single source of truth for visual design.

No component should be introduced unless:

- It solves a real problem.
 - Existing components cannot be reused.
 - Documentation is updated.
 - Design review is completed.
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Related Documents

- 02_Product_Requirements_Document.md
 - 08_Information_Architecture.md
 - 10_Database_Architecture.md
 - 11_API_Architecture.md
 - 12_Engineering_Standards.md
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Approval Status

Version: 1.0.0

Status: Draft

End of Document